MAG元素循环基因谱

砷 氮 硫 铁

CK组 A组 B组

DYIB01 sp. [63]
Ustatibacter sp. [48]
Ustatibacter sp. [22]
PNJM01 sp013821865 [30]
DASWUD01 sp. [76]
Thiobacteraceae (f) [34]
2-02-FULL-66-14 sp004298295 [36]
Thiobacillus sp. [49]
DSOT01 sp. [123]
UBA6901 sp. [86]
Sulfuricaulis sp. [141]
SCGC-AG-212-J23 sp036524215 [88]
SCGC-AG-212-J23 sp. [127]
Pseudolabrys sp. [95]
Sulfuricaulis sp. [110]
Sulfuritortus sp. [70]
JADGHX01 sp. [136]
Gallionella sp. [152]
Gallionella sp. [94]
Nitrosomonas sp. [69]
Gallionella sp. [90]
CADEED01 sp. [125]
SCGC-AG-212-J23 sp. [128]
SZUA-430 sp. [14]
Rhizomicrobium sp. [151]
Ga0077527 sp. [71]
Dongiaceae (f) [101]
JACPFY01 sp. [116]
CADEEJ01 sp. [27]
FEN-1219 sp. [142]
Vitreimonas sp. [21]
CADEFF01 sp. [39]
UBA2999 (f) [78]
JADGNW01 sp. [87]
UBA6911 sp. [133]
JACPDI01 sp. [156]
JACPDH01 sp. [133]
DATLYG01 sp. [25]
Palsa-187 sp. [38]
JAYUPY01 sp. [114]
JACPDH01 sp. [15]
UBA6911 sp. [150]
Koribacteraceae (f) [165]
DASYVL01 sp. [173]
JACQNR01 sp. [18]
Gp6-AA40 sp. [72]
JAICEW01 sp. [139]
DATMXP01 sp035561515 [172]
Gp7-AA8 (f) [118]
Koribacteraceae (f) [137]
UBA5066 sp. [52]
DASTQD01 sp. [67]
UBA5704 sp. [130]
DATGMB01 sp. [160]
JACTMD01 sp. [43]
QHYT01 sp. [75]
DASYVL01 sp. [82]
JAJPHL01 (f) [126]
JAKEFK01 sp. [26]
DATKME01 sp. [3]
Palsa-187 sp. [42]
Deflavilinea sp. [140]
JADZEK01 sp. [117]
Deflavilinea sp. [124]
JAKOUW01 sp. [154]
Deflavilinea sp. [20]
Fen-1038 sp016195815 [102]
Flexifilum sp. [161]
JACRMK01 sp. [60]
JAKALX01 sp. [145]
Villigracilaceae (f) [166]
SPCO01 sp016200745 [98]
SPCO01 sp. [45]
UBA6663 (f) [53]
UBA5189 sp. [111]
Flexifilum sp. [148]
SPCO01 sp. [112]
WHTK01 sp. [132]
Deflavilinea sp. [147]
UBA5189 sp. [33]
UBA5189 sp. [53]
UBA5189 sp. [83]
JAMCZK01 sp. [93]
DSTU01 sp. [40]
UBA5189 sp. [96]
UBA5189 sp. [2]
JAGOBP01 sp. [74]
Deflavilinea sp. [77]
UBA11986 sp. [68]
Fen-1258 sp. [120]
JACPLA01 sp. [12]
JAERML01 sp. [13]
CAADGV01 sp. [84]
DATFVA01 sp. [9]
JACPLA01 sp. [41]
RPQH01 sp. [50]
DASXTZ01 sp. [10]
OLB6 sp. [7]
GWC2-37-13 sp. [51]
DATKGN01 sp. [62]
C7867-001 sp. [104]
DATIGJ01 sp. [131]
JAMDEL01 sp. [17]
DASUOD01 sp. [65]
2-01-FULL-50-25 sp. [91]
2-01-FULL-51-34 sp. [164]
DATGWK01 (f) [149]
JANWTN01 (f) [46]
JACPFJ01 sp. [99]
SM23-61 (f) [109]
Unclassified [157]
YEO1/01 sp. [28]
Geoanaerobacter sp009684525 [32]
Geobacter sp. [103]
UBA5619 (f) [19]
UBA5619 sp. [106]
CAIXSE01 sp. [105]
Cellulomonas sp. [115]
JAEZGP01 sp. [167]
BP-133 sp. [168]
JANNHK01 sp. [99]
VAYN01 sp. [59]
VFJN01 sp. [80]
CAJCYJ01 (f) [107]
JADJOM01 sp. [1]
JACDDX01 sp. [129]
DASXPG01 sp. [6]
Gemmatimonadaceae (f) [138]
Gemmatimonadaceae (f) [79]
SCN-70-22 sp027146425 [8]
DASPMP01 sp. [37]
JAAXZI01 sp. [111]
JACPRB01 (f) [153]
JAPLNB01 sp. [81]
JABWBB01 sp. [29]
JAEUJZ01 sp. [121]
JADLGW01 sp. [73]
Anaeromyxobacter sp. [119]
JADKCZ01 sp. [5]
JACRMN01 sp. [66]
Palsa-1104 (f) [100]
Vulgatibacteraceae (f) [4]
Nitrospira_A sp030123565 [122]
Nitrospira_D sp. [56]
UBA9217 (f) [134]
JAAYUK01 sp. [61]
JACQBZ01 (f) [144]
FEB-12 sp. [44]
JADJGC01 sp. [97]
DSPP01 sp. [159]
Didemnitutus sp. [54]
JBBRWD01 sp. [170]
UBA8199 (f) [85]
AR12 sp. [11]
2-02-FULL-66-22 sp. [108]
UBA5197 sp. [155]
UBA5572 sp. [145]
DP-19 sp. [92]
DP-20 sp. [158]
Deferrimicrobium sp. [23]
JAENXD01 sp. [58]
Nitrosotenuis sp. [64]
UBA8941 sp. [31]
Methanoregula sp. [16]
Elainellaceae (f) [163]
JAWXBZ01 (f) [24]
JACRMM01 sp. [35]
Ga0074140 sp. [57]

门水平
Pseudomonadota
Acidobacteriota
Chloroflexota
Bacteroidota_A
Patescibacteria
Desulfobacterota
Actinomycetota
Gemmatimonadota
Planctomycetota
Myxococcota
Nitrospinota
Zixibacteria
Verrucomicrobiota
Methylomirabilota
Bacteroidota
Desulfobacterota_B
Desulfobacterota_E
Thermoproteota
Halobacteriota
Cyanobacteriota
Elusimicrobiota
Desulfobacterota_C
Chlamydiota

基因拷贝数
砷
氮
硫
铁

行标注
丰度排名
关键物种
两者兼具

组平均丰度
>0.50
0.30~0.50
0.20~0.30
0.15~0.20
0.10~0.15
0.05~0.10
0.01~0.05
<0.01

实验分组
A组
B组

功能分类
砷酸盐还原
亚砷酸盐氧化
呼吸型砷酸盐还原
砷转运与解毒
砷甲基化
硝调控
硝酸盐还原
亚硝酸盐还原
NO还原
N2O还原
氨氧化
固氮
硫酸盐同化还原
硫酸盐异化还原
硫化物氧化
硫代硫酸盐代谢
铁转运系统
铁摄取调控

arsC-grx *arsC-trx* *arsC-mix* *aoxA* *aoxB* *arsB* *ACR3* *arsA* *arsH* *ASMT* *arsR* *narG* *narH* *narI* *napA* *napB* *narB* *nirK* *nirS* *nirA* *CYP55* *norB* *nosZ* *nnoA* *nmoB* *nmoC* *nifD* *nifK* *cysJ* *cysI* *cysH* *sir* *aprA* *aprB* *dsrA* *dsrB* *sqr* *soxA* *soxX* *soxB* *soxY* *soxZ* *TST* *ABC.FEV.A* *TC.FEV.OM* *ABC.FEV.P* *ABC.FES.S* *afuA* *FTR* *fur* *tonB*